

# Castalia

A four-channel electrochemical sensing SoC with a five-hart RISC-V back end

Maxx Seminario

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# Where cancers are caught, and whether anyone looks

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Target  
application

The chip

Analog front end

Operating modes

Analog blocks

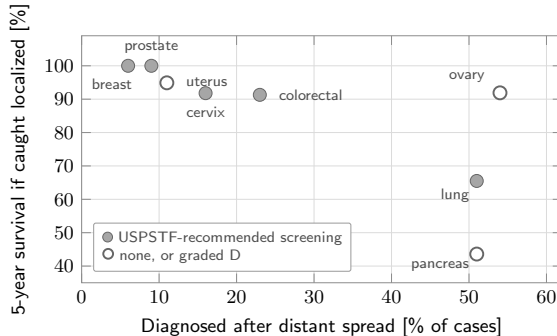
Digital

Against Myshkin  
and the field

From currents to  
a score

References

Backup



[1] SEER 21, 2016–2022; [2, 3] USPSTF

# Why ovarian: late, lethal, and unscreened

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| Site       | Distant at dx | Deaths/100 cases | Screening (USPSTF) |
|------------|---------------|------------------|--------------------|
| Ovary      | 54 %          | 59               | D                  |
| Pancreas   | 51 %          | 78               | D                  |
| Lung       | 51 %          | 54               | B, LDCT            |
| Colorectal | 23 %          | 35               | A                  |
| Cervix     | 16 %          | 31               | A                  |
| Uterus     | 11 %          | 21               | none               |
| Prostate   | 9 %           | 11               | C, PSA             |
| Breast     | 6 %           | 13               | B                  |

USPSTF: A/B recommended; C offered selectively; D recommended *against*; none: no statement [7].

[1] SEER, 2026 estimates; [2, 3] USPSTF

Most lethal gynecologic cancer [1]: 92 % survivable if caught localized, yet 54 % present after spread, and screening is graded D [2]. UKCTOCS moved stage, not mortality [4].

# One serum aliquot, four sites, one score

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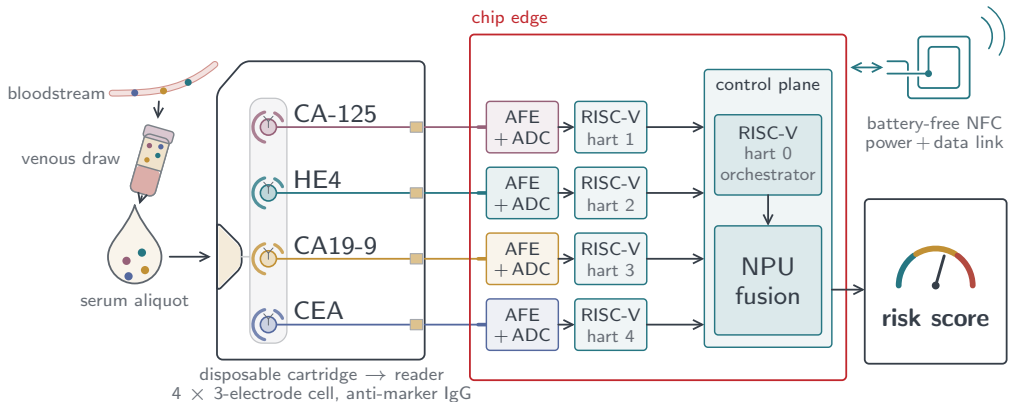
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**Symptomatic-patient triage on a disposable four-site cartridge; four markers read at the same instant, fused on chip.**

# No single marker is a test; four fused are [5]

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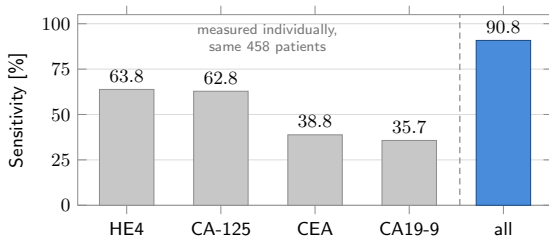
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| Marker | Molecule                          | Cut-off   |
|--------|-----------------------------------|-----------|
| CA-125 | MUC16 mucin epitope [8]           | 35 U/mL   |
| HE4    | WFDC2 glycoprotein [6]            | 70 pmol/L |
| CA19-9 | sialyl-Lewis <sup>a</sup> antigen | 37 U/mL   |
| CEA    | CEACAM5 glycoprotein              | 5 ng/mL   |

Each marker covers a subtype the others miss. The decision is a learned combination, so the chip is four channels plus the processor that fuses them.

# Block diagram

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Four channel tiles own four AFE sites; every master reaches ROM, RAM, NPU and peripherals only through the arbiter.

# Physical implementation

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**ARTWORK PENDING**



**ARTWORK  
PENDING**



# Grounded-WE two-amplifier potentiostat

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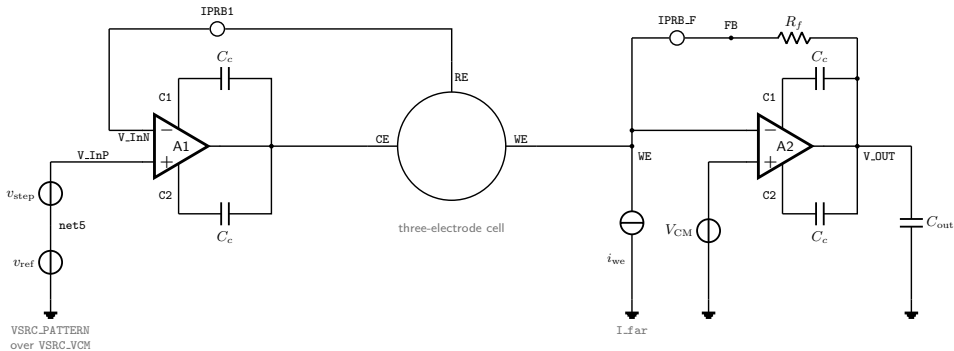
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**A1 drives CE with RE in feedback; A2 holds WE at  $v_{cm}$  through  $R_f$ .  $v_{out} = v_{cm} - I_{we}R_f$ , both polarities.**

# Analog top level

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**ARTWORK PENDING**



Virtuoso export of the analog top level: four channels, shared bias, test bus

# The electrode model, in Verilog-A

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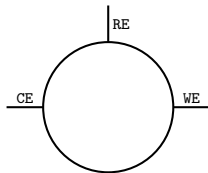
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$R_u$ ,  $R_{ce}$ ,  $C_{dl}$  lumped; the faradaic branch  $Z_f$  is the Verilog-A element.

$i_f$  faradaic current  
 $\eta$  overpotential,  $E - E^{0'}$   
 $f$   $F/RT$   
 $C_{O,R}(0, t)$  surface conc. of O, R  
 $C^*$  bulk conc.  
 $\delta$  Nernst layer

Kinetics  $i_f = nFAk^0 [C_R(0, t) e^{(1-\alpha)nf\eta} - C_O(0, t) e^{-\alpha nf\eta}]$   
 Transport  $\partial C / \partial t = D \partial^2 C / \partial x^2$ , finite-volume, 28 expanding slabs per species  
 Boundaries flux =  $\pm i_f / nFA$  at the surface;  $C(\delta, t) = C^*$  at the Nernst layer

| Parameter | Symbol   | Default             | Unit                        | Quantity                             |
|-----------|----------|---------------------|-----------------------------|--------------------------------------|
| E0        | $E^{0'}$ | 0.0                 | V                           | formal potential of the couple       |
| nel       | $n$      | 1                   | —                           | electrons transferred                |
| alpha     | $\alpha$ | 0.5                 | —                           | transfer coefficient                 |
| k0        | $k^0$    | 0.1                 | $\text{cm s}^{-1}$          | standard heterogeneous rate constant |
| Dif       | $D$      | $5 \times 10^{-6}$  | $\text{cm}^2 \text{s}^{-1}$ | diffusion coefficient, both species  |
| Area      | $A$      | 0.01                | $\text{cm}^2$               | electrode area                       |
| Cox       | $C_O^*$  | $1 \times 10^{-6}$  | $\text{mol cm}^{-3}$        | bulk concentration, oxidised species |
| Cred      | $C_R^*$  | 0                   | $\text{mol cm}^{-3}$        | bulk concentration, reduced species  |
| Temp      | $T$      | 298.15              | K                           | interface temperature                |
| expmax    | —        | 40                  | —                           | exponent clamp on the kinetic terms  |
| gpar      | —        | $1 \times 10^{-15}$ | S                           | DC shunt across the terminals        |

Equations: Bard–Faulkner [13]. Discretisation: Britz–Strutwolf, Feldberg [17, 18]. Precedent for Verilog-A electrode co-simulation: Jog et al. [19].

# Emulating the four markers

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| Marker | Reporter       | $E^{0'}$ (V) | Peak (nA) | $C_R^*$ ( $\mu$ M) | Cut-off         |
|--------|----------------|--------------|-----------|--------------------|-----------------|
| CA-125 | methylene blue | -0.15        | 856       | 279                | 35 U/mL [5, 8]  |
| HE4    | ferrocene      | +0.10        | 1311      | 394                | 70 pmol/L [6]   |
| CA19-9 | TMB / HRP [20] | +0.25        | 1659      | 458                | 37 U/mL [5]     |
| CEA    | anthraquinone  | +0.40        | 939       | 303                | 5 ng/mL [5, 14] |

$E^{0'}$  reporter formal potential, i.e. peak position; chosen in the  $-0.6$  to  $+0.4$  V window [22, 23], 150–250 mV apart, not measured per reporter

Peak inside 50 nA to 5  $\mu$ A on screen-printed electrodes [14, 15, 21];  $C_R^*$  fitted to reach it

$k^0$  0.1 cm/s, reversible; nothing published on carbon for these markers

Shared  $\alpha = 0.5$ ,  $n = 1$ ,  $D = 5 \times 10^{-6}$  cm<sup>2</sup>/s,  $A = 0.01$  cm<sup>2</sup>, 37 °C,  $R_f = 560$  k $\Omega$

**Representative values, not marker-measured: these slides demonstrate the channel, not an assay.**

# Channel range, assay range, and where the panel was placed

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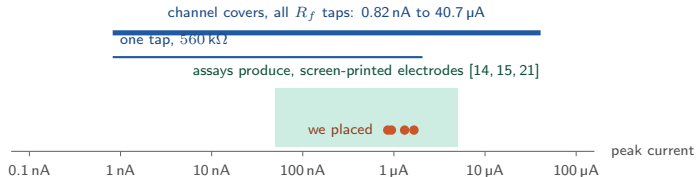
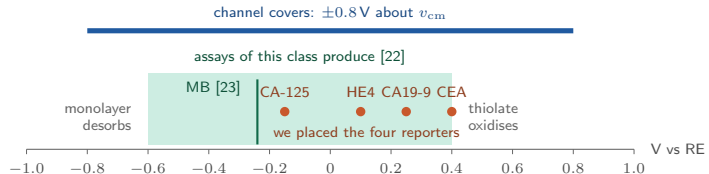
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Potentials: spaced 150–250 mV for one-sweep resolution inside the reporter window, not per-reporter values. Peak heights: inside the published envelope and one tap's full scale, then  $C_R^*$  fitted. Silicon is unaffected: the channel covers the assay band with 0.2–0.4 V and a decade of current to spare, so moving the dots changes what the simulation shows, not what the chip can measure.

# Exercising the cell: differential-pulse voltammetry

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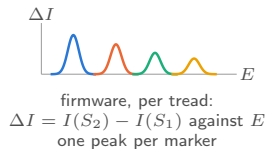
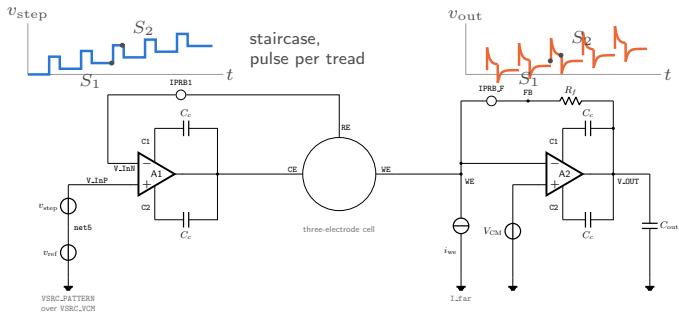
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Excite

DAC: staircase with a pulse on every tread, applied at RE by A1

Sense

$$v_{out} = v_{cm} - i_{WE} R_f$$

Sample

SAR at  $S_1$  (before the pulse) and  $S_2$  (end of the pulse), once charging has decayed

Resolve

firmware:  $\Delta I = I(S_2) - I(S_1)$  vs  $E$ ; peak position: marker, peak height: concentration

# Four markers resolved in one differential-pulse sweep

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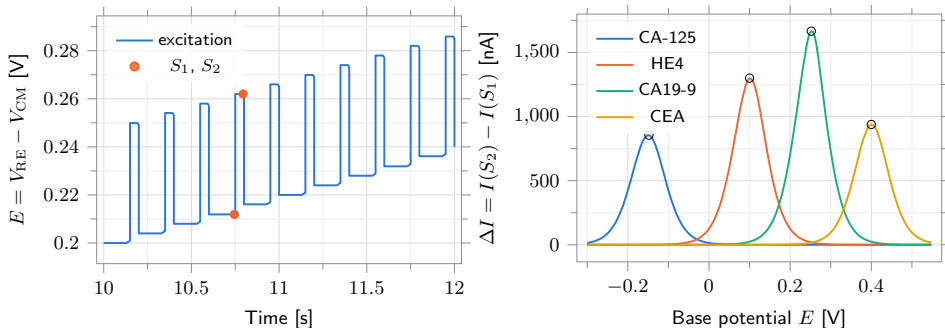
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\*schematic sim

Peak position identifies the reporter, peak height carries concentration. Apexes within 2 mV and 1 % of the assay targets; blank below 1 pA.

# Every clinical cut-off lands inside the channel window

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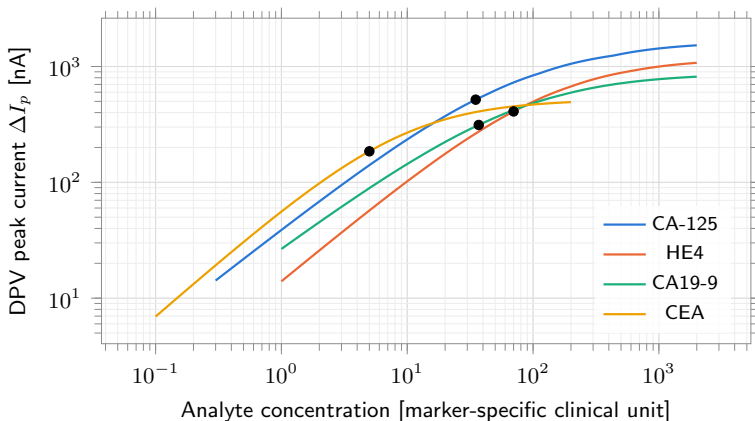
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\*schematic sim

CA-125 35 U/mL, HE4 70 pmol/L, CA19-9 37 U/mL, CEA 5 ng/mL: all four on the steep part of the curve at one  $R_f$  setting.

# Cyclic voltammetry through the assembled channel

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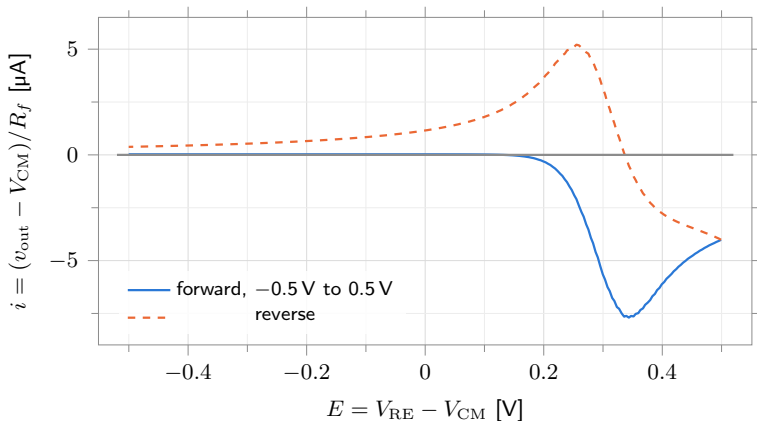
Digital

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\*schematic sim

Staircase potential from the on-chip DAC, current digitised by the on-chip SAR, unclipped over the sweep.

# How impedance is measured: the same loop, no extra hardware

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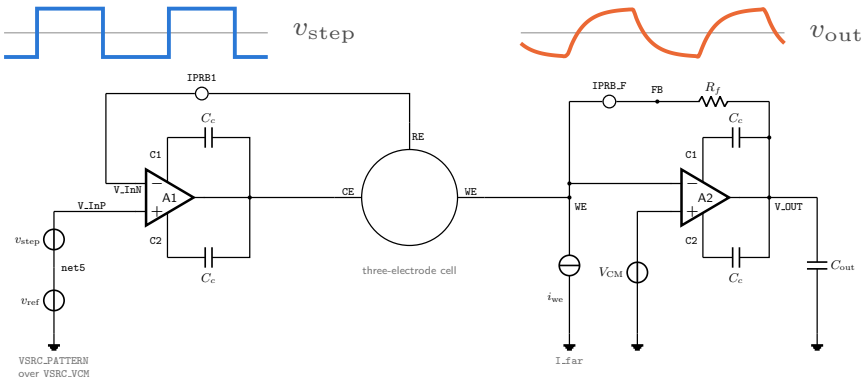
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Excite  
Sense  
Digitise  
Solve

DAC: a small two-code square about the DC setpoint  $v_{ref}$   
 $v_P$  across the cell;  $i_{WE} = -v_{out}/R_f$   
 the SAR records both; firmware I/Q demodulation  
 $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$ , or a ratio to  $Z_{cal}$

# Magnitude and phase from codes: I/Q demodulation

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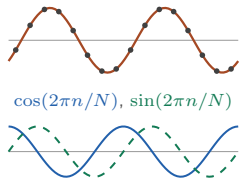
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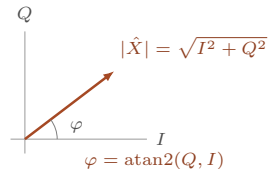
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record  $c_n$ ,  $N$  per period,  $M$  periods



$$I = \frac{2}{MN} \sum c_n \cos \frac{2\pi n}{N}$$
$$Q = \frac{2}{MN} \sum c_n \sin \frac{2\pi n}{N}$$



Records  $v_P$  and  $v_{out}$   
Divide  $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$   
Ratio against  $Z_{cal}$ :  $R_f$  drops out

# The four loads behind the EIS plots

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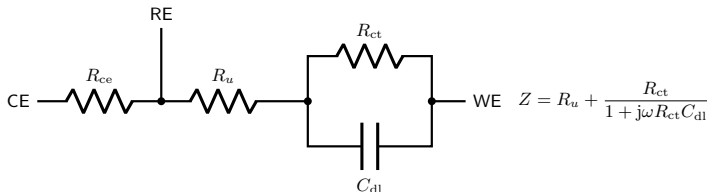
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| Load     | $R_{ct}$       | $C_{dl}$   | Gain code | Stands for                                                                                            |
|----------|----------------|------------|-----------|-------------------------------------------------------------------------------------------------------|
| resistor | –              | –          | $N = 0$   | an 11 k $\Omega$ resistor in place of the cell: no interface, so the trace is the channel's own error |
| fouled   | 10 k $\Omega$  | 10 $\mu$ F | $N = 0$   | a degraded electrode: adsorbed protein and biofilm open leakage paths and enlarge the interface       |
| typical  | 100 k $\Omega$ | 1 $\mu$ F  | $N = 20$  | an electrode in use                                                                                   |
| fresh    | 1 M $\Omega$   | 100 nF     | $N = 63$  | a new, clean surface                                                                                  |

$R_u = R_{ce} = 1 \text{ k}\Omega$  throughout. Fouling:  $R_{ct}$  falls and  $C_{dl}$  rises through an electrode's life; tracking that drift is what EIS is for here.

# Impedance spectroscopy on the same channel

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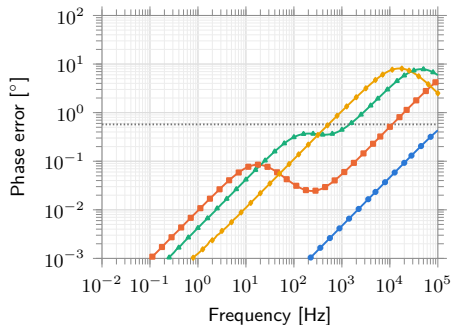
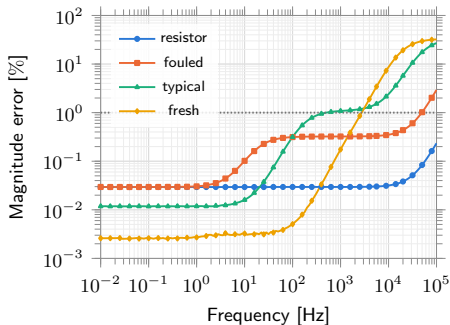
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Accuracy  $\pm 1\%$ ,  $\pm 0.6^\circ$  to 5 kHz;  $\pm 5\%$  to 27 kHz. Limit: A1's loop bandwidth  
 Range 246  $\Omega$  to 12.1 M $\Omega$ . Limits: TIA compliance below, one LSB of output above; 320  $R_f$  codes in between

\*schematic sim

The excitation amplitude never enters:  $Z$  is the ratio of two records through one converter, so DAC gain and droop cancel.

# What sets the EIS floor, and what averaging buys

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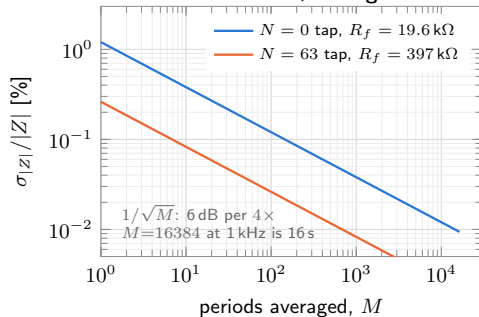
**Systematic floor: A2's finite gain, scaled by the noise gain**

$$\varepsilon_{\text{dc}} \approx \frac{1 + R_f/|Z|}{A_{\text{OL}}}$$

| Load     | $ Z _{\text{dc}}$ | $R_f$           | $R_f/ Z $ | Error    |
|----------|-------------------|-----------------|-----------|----------|
| resistor | 11 k $\Omega$     | 19.6 k $\Omega$ | 1.8       | 0.029 %  |
| fouled   | 11 k $\Omega$     | 19.6 k $\Omega$ | 1.8       | 0.030 %  |
| typical  | 101 k $\Omega$    | 140 k $\Omega$  | 1.4       | 0.003 %  |
| fresh    | 1 M $\Omega$      | 397 k $\Omega$  | 0.4       | 0.0007 % |

Set by the gain code, not the electrode; repeatable, so the  $Z_{\text{cal}}$  ratio removes it (next slide). What is left is the random term on the right.

**Random error: front-end noise, averaged**



**Bias is set by the gain code and cancels in the ratio; precision is bought with measurement time,  $\sigma \propto 1/\sqrt{M}$ .**

# Calibrating impedance: what cancels, and what a 20 % poly costs

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$$\hat{Z} = Z_{\text{cal}} \frac{(\hat{V}_P / \hat{V}_{\text{OUT}})}{(\hat{V}_P / \hat{V}_{\text{OUT}})_{\text{cal}}} \quad \text{same gain code, same frequency, same converter}$$

|             | On-chip $Z_{\text{cal}}$ (per decade, muxed in)                                 | Off-chip standard (at test)          |
|-------------|---------------------------------------------------------------------------------|--------------------------------------|
| Cancels     | $R_f$ , its pole, the DAC, the LSB, A2's gain floor                             | the same                             |
| Value known | to the poly's chip-to-chip spread, $\sim 20\%$                                  | to the standard, $< 0.1\%$           |
| When        | any time, in the field, before every point                                      | once per chip, stored in flash       |
| Tracks      | temperature and ageing, since $Z_{\text{cal}}$ and $R_f$ are the same poly      | the channel on the day it was tested |
| Good for    | trends: fouling is a change in one electrode's $Z$ , untouched by a fixed scale | absolute ohms                        |

**The poly need not be accurate, only stable. One off-chip number per chip pins the absolute scale; the on-chip ratio carries it through temperature and time.**

# The class-AB amplifier: one cell, both roles

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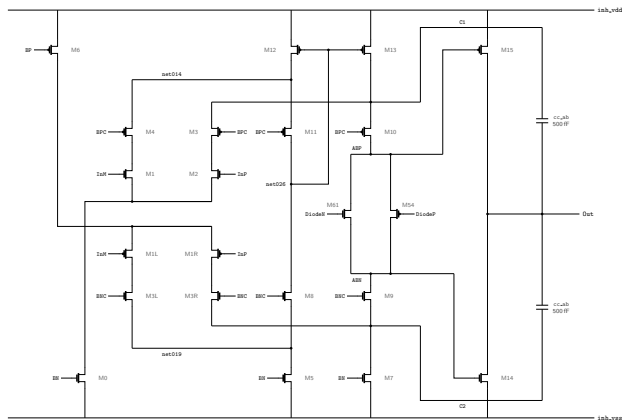
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complementary  
input pairs and  
their cascodes

diode-connected  
summing branch

class-AB  
drive

output

testbench  
compensation

# Bipolar accuracy through zero

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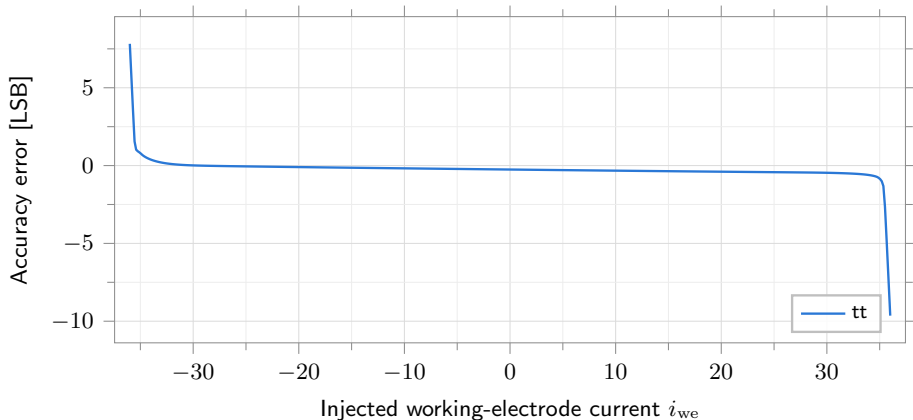
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\*schematic sim

**No kink at the crossing: class-AB output, no dead zone. This is the measurement rev-1's PMOS mirror could not make.**

# Programmable feedback resistor: one string, 320 codes

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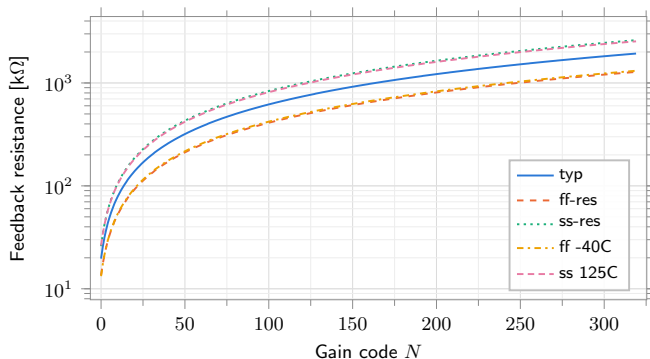
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$$R_f = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$$

| $N$        | $R_f$   | Full scale |
|------------|---------|------------|
| 0          | 19.6 kΩ | ±40.7 μA   |
| 63         | 397 kΩ  | ±2.02 μA   |
| 319        | 1.93 MΩ | ±0.41 μA   |
| LSB at 319 |         | 0.824 nA   |

Strictly monotonic at every corner, including the four thermometer seams. Maximum gain is noise-limited, not quantisation-limited.

# 10-bit SAR ADC, extracted netlist

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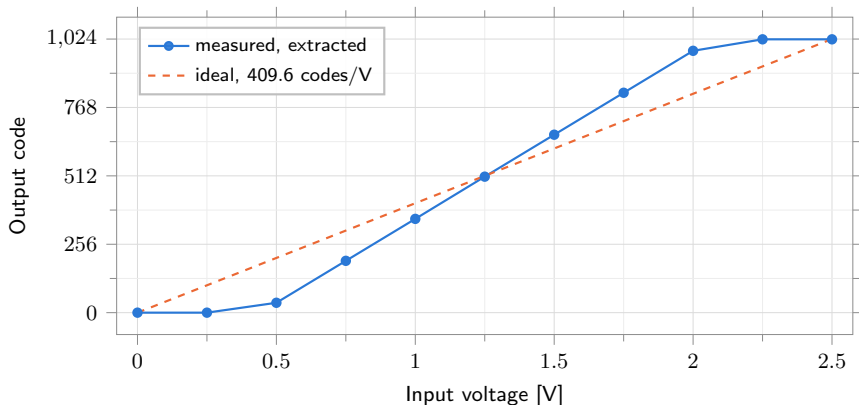
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■ extracted

netlist

**23 clocks per conversion, 870 kSa/s, 50 pJ. The capacitor array exists only in layout, so this is the one block characterised extracted.**

# Block results: corners, with mismatch $\sigma$

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| Parameter                                         | Unit       | tt       | ff       | fs       | sf       | ss       | Spread | $\sigma$ (MC) |
|---------------------------------------------------|------------|----------|----------|----------|----------|----------|--------|---------------|
| <b>Class-AB amplifier (A1 and A2)</b>             |            |          |          |          |          |          |        |               |
| Open-loop gain                                    | dB         | 82.77    | 79.88    | 83.14    | 83.37    | 84.94    | 5.06   | 1.06          |
| Phase margin                                      | deg        | 73.52    | 74.70    | 74.04    | 72.86    | 73.34    | 1.84   | 1.42          |
| Dominant pole                                     | kHz        | 1.592    | 2.286    | 1.538    | 1.494    | 1.225    | 1.061  | 0.276         |
| Gain-bandwidth product                            | MHz        | 22.78    | 23.19    | 22.89    | 22.94    | 22.59    | 0.60   | 3.57          |
| Input offset voltage                              | mV         | 0.622    | 0.649    | 0.624    | 0.569    | 0.456    | 0.193  | 5.201         |
| <b>Programmable feedback resistor<sup>a</sup></b> |            |          |          |          |          |          |        |               |
| $R$ at $N=0$                                      | k $\Omega$ | 19.64    | —        | —        | —        | —        | —      | 0.071         |
| $R$ at $N=319$                                    | M $\Omega$ | 1.931    | —        | —        | —        | —        | —      | 0.0007        |
| Seam step 63 $\rightarrow$ 64                     | k $\Omega$ | 7.42     | —        | —        | —        | —        | —      | 0.504         |
| <b>10-bit SAR ADC, extracted<sup>b</sup></b>      |            |          |          |          |          |          |        |               |
| Fitted slope                                      | codes/V    | 628.59   | 628.59   | 628.40   | 628.73   | 628.13   | 0.60   | —             |
| Effective LSB                                     | mV         | 1.5909   | 1.5909   | 1.5914   | 1.5905   | 1.5920   | 0.0015 | —             |
| Input at code 0                                   | V          | 0.4408   | 0.4408   | 0.4407   | 0.4408   | 0.4399   | 0.0009 | —             |
| Usable span                                       | V          | 1.6274   | 1.6274   | 1.6280   | 1.6271   | 1.6286   | 0.0015 | —             |
| Worst residual                                    | LSB        | 0.495    | 0.495    | 0.490    | 0.477    | 0.484    | 0.018  | —             |
| RMS residual                                      | LSB        | 0.294    | 0.294    | 0.277    | 0.288    | 0.302    | 0.025  | —             |
| Analog supply current                             | $\mu$ A    | 6.36     | 4.79     | 5.45     | 8.44     | 15.43    | 10.64  | —             |
| Energy per conversion                             | pJ         | 20.8     | 19.3     | 18.1     | 27.3     | 49.5     | 31.4   | —             |
| <b>12-bit R-2R reference DAC, ideal rail</b>      |            |          |          |          |          |          |        |               |
| Full-scale output                                 | V          | 2.499388 | 2.499388 | 2.499388 | 2.499388 | 2.499388 | 0      | 0.000001      |
| LSB size                                          | $\mu$ V    | 610.351  | 610.351  | 610.351  | 610.351  | 610.351  | 0      | 0.0003        |
| INL, most positive                                | LSB        | 0.113    | 0.101    | 0.100    | 0.127    | 0.130    | 0.030  | 0.610         |
| INL, most negative                                | LSB        | -0.376   | -0.338   | -0.390   | -0.362   | -0.414   | 0.076  | 0.620         |
| DNL, most positive                                | LSB        | 0.002    | 0.001    | 0.002    | 0.002    | 0.002    | 0.000  | 1.128         |
| DNL, most negative                                | LSB        | -0.423   | -0.374   | -0.424   | -0.424   | -0.479   | 0.105  | 1.415         |

$\sigma$ : 200 samples, process + mismatch, nominal corner. Corner runs hold the passive models at typical, so resistor and ladder spreads are lower bounds. <sup>a</sup>The TRM carries  $R_f$  corners as a figure (the  $R_f$  slide), not a table. <sup>b</sup>Extracted netlist; the extraction carries no mismatch statistics, so no  $\sigma$ . Amplifier and DAC: schematic sim\*; SAR: extracted.

# One AFE channel, as simulated

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| Parameter              | Value                                                                  | Condition                                        |
|------------------------|------------------------------------------------------------------------|--------------------------------------------------|
| CE compliance          | 13.2 mV – 2.480 V                                                      | 2.5 V rail, corners                              |
| Cell current delivered | $\pm 52.5 \mu\text{A}$                                                 | 12 k $\Omega$ cell                               |
| TIA compliance         | $\pm 0.80 \text{ V}$ about $v_{\text{cm}}$                             |                                                  |
| Feedback resistor      | 19.6 k $\Omega$ – 1.93 M $\Omega$ , 320 codes                          | monotonic at every corner                        |
| Full-scale current     | $\pm 40.7 \mu\text{A}$ ( $N=0$ ) to $\pm 0.41 \mu\text{A}$ ( $N=319$ ) | SAR window $\cap$ TIA compliance                 |
| Current quantum        | 0.824 nA at maximum gain                                               | below the 2 nA noise budget                      |
| Bipolar accuracy       | 0.25–0.38 LSB worst                                                    | 560 k $\Omega$ , 1 M $\Omega$ electrode, corners |
| Control-loop PM        | 79–80°                                                                 | five corners, into a cell                        |
| TIA-loop PM            | 89–93°                                                                 | 12-cell electrode grid                           |
| Input current noise    | 341 pA standalone, 3.6–21 nA into an electrode                         | 0.1–100 Hz, 560 k $\Omega$                       |
| Zero-current offset    | $\sigma(v_{\text{re}}) = 5.72 \text{ mV}$ , gain-independent           | 200 MC samples, calibratable                     |
| Quiescent current      | 197–250 $\mu\text{A}$ per channel                                      | amplifiers + local bias                          |

\*schematic sim    ■ Monte Carlo, 200 samples

Pre-layout, electrode held nominal: no row includes electrode-to-electrode spread.

# Castalia at a glance

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## Digital

|                 |                                        |
|-----------------|----------------------------------------|
| Harts           | 5 (1 orchestrator, 4 channel tiles)    |
| ISA             | rv32imac_zba_zbb_zbs_zbc               |
| Clock           | 24 MHz rated, closed at 25 MHz         |
| Measured CPI    | 1.260 aggregate, nine benchmarks       |
| Private TCM     | 8 KiB per hart, 1-cycle                |
| Shared RAM      | 80 KiB behind a round-robin arbiter    |
| Synchronisation | 16 hardware mutexes, LR/SC, AMOs       |
| Interrupts      | claim/complete, any vector to any hart |
| Accelerator     | NPU: MLP, conv-1D, XNOR, GEMM          |
| Power domains   | 4 MTCMOS, one per channel tile         |
| Debug           | JTAG, RISC-V Debug Module              |
| Field boot      | NFC ISO 14443A harvested boot          |

## Analog and physical

|                   |                                                                |
|-------------------|----------------------------------------------------------------|
| Measurement sites | 4 identical, simultaneous                                      |
| Topology          | bipolar grounded-WE potentiostat                               |
| Electrode pads    | 12 (CE/RE/WE $\times$ 4)                                       |
| Converter         | 10-bit SAR per channel, extracted                              |
| Reference DAC     | 12-bit R-2R per channel                                        |
| Feedback resistor | $18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$ , 320 codes     |
| Full scale        | $\pm 40.7\text{ }\mu\text{A}$ to $\pm 0.41\text{ }\mu\text{A}$ |
| EIS               | firmware mode of the same channels                             |
| Analog supply     | $2.5\text{ V}$ , $v_{\text{cm}} = 1.25\text{ V}$               |
| Process           | 65 nm CMOS, 1P8M, HVT cells                                    |
| Die / package     | $3.0 \times 3.0\text{ mm}$ / LQFP-100                          |
| Digital power     | 10.72 mW post-route, SS, 25 MHz                                |

Digital numbers from the hardened flow; analog numbers schematic-level, except the SAR, which is extracted.

# Digital, as closed

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## Whole chip, post-route

|           |         |
|-----------|---------|
| Internal  | 3.78 mW |
| Switching | 2.33 mW |
| Leakage   | 4.61 mW |

|              |                 |
|--------------|-----------------|
| <b>Total</b> | <b>10.72 mW</b> |
|--------------|-----------------|

## One channel tile

|       |          |
|-------|----------|
| Total | 1.761 mW |
| Gated | 0        |

## Measured CPI, timed kernels

| Benchmark        | Instr.           | CPI          |
|------------------|------------------|--------------|
| spmv             | 802 447          | 1.143        |
| multiply         | 20 892           | 1.163        |
| vvadd            | 2 408            | 1.374        |
| median           | 4 015            | 1.398        |
| qsort            | 123 205          | 1.398        |
| rsort            | 146 548          | 1.412        |
| dhrystone        | 200 523          | 1.504        |
| towers           | 3 747            | 1.682        |
| memcpy           | 11 017           | 1.726        |
| <b>Aggregate</b> | <b>1 314 802</b> | <b>1.260</b> |

## Timing at 25 MHz

|               |        |
|---------------|--------|
| reg2reg WNS   | +99 ps |
| Leakage share | 43 %   |

■ hardened flow, post-route

SS 0.9 V 125 °C, statistical activity 0.2. Gating one tile removes 1.76 mW; the always-on band pays nothing.

# Power gating: one switched rail per channel tile

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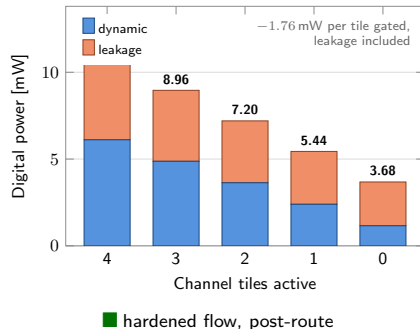
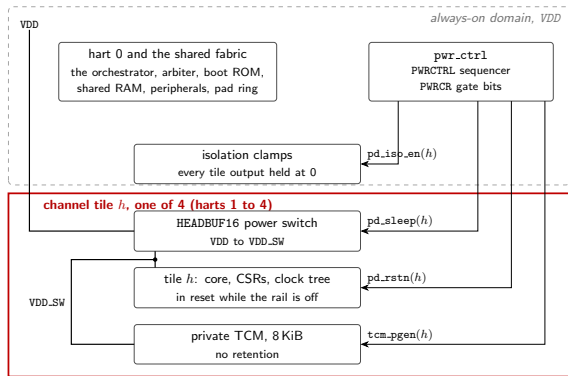
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**Cold gating:** the header cuts the rail, so a parked tile draws neither dynamic nor leakage power. Each tile gated removes 1.76 mW; the always-on band keeps 3.68 mW.

# Against the field: the ISCAS'26 table, plus Castalia

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| Parameter         | Castalia (this work)   | Myshkin [9]       | Chen '19 [10] | Ghoreishizadeh '17 [11] | Lin '23 [12] |
|-------------------|------------------------|-------------------|---------------|-------------------------|--------------|
| Modality          | <b>CA/CV/DPV/EIS</b>   | CA/CV             | CA/CV         | CA/CV                   | CV/EIS       |
| Channels          | <b>4, simultaneous</b> | 1                 | 1             | 1, differential         | 1            |
| Current direction | <b>bipolar</b>         | unipolar          | bipolar       | bipolar                 | bipolar      |
| Sensitivity (pA)  | 341 <sup>a</sup>       | 56.5 <sup>b</sup> | 41            | 0.47                    | 1.7          |
| $I_{\max}$ (μA)   | ±40.7                  | +20               | ±5            | ±20                     | -48/+16      |
| Power (mW)        | 3.68 <sup>c</sup>      | 3.0 <sup>d</sup>  | 0.016         | 9.3                     | 1.0          |
| Process (nm)      | 65                     | 65                | 180           | 350                     | 180          |
| On-chip processor | <b>5 RISC-V harts</b>  | 1 RISC-V hart     | none          | none                    | none         |
| On-chip ML        | <b>NPU, 4 modes</b>    | NPU, 1 mode       | none          | none                    | none         |

Castalia column is best case and pre-silicon. <sup>a</sup>AFE only: input-referred rms, 0.1–100 Hz, maximum gain, ideal  $R_f$ , no electrode; the converter LSB is 0.82 nA. <sup>b</sup>AFE only; ADC quantisation limits the system to 153 pA. <sup>c</sup>Digital only: hart 0 awake, all four channel tiles gated, post-route SS 125 °C 25 MHz; one channel measuring adds 1.76 mW (tile) + 1.14 mW (AFE). <sup>d</sup>Measured: 2.8 mW digital at 24 MHz + 258 μW AFE.

**Still the only entries with a processor and an ML engine on the die; Castalia adds concurrency, both current polarities and EIS.**

# The fusion runs on the NPU

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$$s = \sigma \left( \sum_{k=1}^4 w_k \hat{c}_k + b \right)$$

|         |                                                   |
|---------|---------------------------------------------------|
| Input   | peak current, peak potential, baseline $\times 4$ |
| Mode    | 0, dense layer, one sigmoid                       |
| Output  | one risk score                                    |
| Weights | trained off-chip, staged by firmware              |

## Per site

|           |                                   |
|-----------|-----------------------------------|
| Hart      | 1, ownership enforced in hardware |
| Potential | own 12-bit DAC                    |
| Converter | own 10-bit SAR                    |
| Power     | own MTCMOS domain                 |

**The logistic form of the cleared two-marker algorithm, extended to four inputs and moved from the central laboratory onto the electrode.**

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# Backup: electrode model validation

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| Scan rate<br>$\text{mV s}^{-1}$ | $i_p$ Randles–Ševčík<br>$\mu\text{A}$ | $i_p$ model<br>$\mu\text{A}$ | Error<br>% | Error, $C_{\text{dl}} \rightarrow 0$<br>% | $\Delta E_p$<br>$\text{mV}$ |
|---------------------------------|---------------------------------------|------------------------------|------------|-------------------------------------------|-----------------------------|
| 25                              | 0.951                                 | 0.952                        | +0.08      | −0.45                                     | 58.78                       |
| 50                              | 1.345                                 | 1.349                        | +0.32      | −0.42                                     | 59.85                       |
| 100                             | 1.902                                 | 1.915                        | +0.67      | −0.38                                     | 60.41                       |
| 200                             | 2.690                                 | 2.721                        | +1.16      | −0.33                                     | 62.33                       |
| 400                             | 3.804                                 | 3.875                        | +1.86      | −0.25                                     | 64.74                       |

Peak current against Randles–Ševčík and  $\Delta E_p$  against Nicholson [13], reversible couple,  $C_{\text{dl}} = 0$  column isolates the faradaic model.

## Backup: USPSTF screening grades

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| Grade    | Certainty of net benefit          | Definition [7]                                                      | Cancer screening [2, 3]                                              |
|----------|-----------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>A</b> | high, substantial                 | Recommends the service. Offer or provide it.                        | cervix (2018), colorectal 50–75 (2021)                               |
| <b>B</b> | high moderate, or moderate        | Recommends the service. Offer or provide it.                        | breast 40–74 (2024), lung LDCT 50–80 (2021), colorectal 45–49 (2021) |
| <b>C</b> | at least moderate, small          | Offer selectively, by professional judgment and patient preference. | prostate PSA 55–69 (2018)                                            |
| <b>D</b> | no net benefit, or harms outweigh | Recommends <i>against</i> the service. Discourage its use.          | <b>ovary (2018)</b> , pancreas (2019)                                |
| <b>I</b> | cannot be determined              | Evidence insufficient, poor, or conflicting.                        | —                                                                    |
| none     | —                                 | No USPSTF statement exists for the condition.                       | uterus                                                               |

Ovarian D is a verdict on a test that was tried, CA-125 plus transvaginal ultrasound [2, 4], not on the absence of one.